

OOP LAB #3

Task#1:

```
/**
 *
 * @author rehan
 */
package com.mycompany.bankaccount;

public class BankAccount {
    private String accountNumber;
    private String accountHolderName;
    private double balance;

    // Default constructor
    public BankAccount() {
        this.accountNumber = "0000";
        this.accountHolderName = "Unknown";
        this.balance = 0.0;
    }

    // Parameterized constructor (2 params)
    public BankAccount(String accountNumber, String accountHolderName) {
        this.accountNumber = accountNumber;
        this.accountHolderName = accountHolderName;
        this.balance = 0.0;
    }

    // Parameterized constructor (3 params)
    public BankAccount(String accountNumber, String accountHolderName, double balance) {
        this.accountNumber = accountNumber;
        this.accountHolderName = accountHolderName;
        this.balance = balance;
    }

    public void displayInfo() {
```

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OutPut:

```
-----< com.mycompany:BankAccount >-----
[ ] Building BankAccount 1.0-SNAPSHOT
    from pom.xml
    -----[ jar ]-----
[ ] --- resources:3.3.1:resources (default-resources) @ BankAccount ---
    skip non existing resourceDirectory /home/rehan/NetBeansProj
[ ] --- compiler:3.13.0:compile (default-compile) @ BankAccount ---
    Nothing to compile - all classes are up to date.
[ ] --- exec:3.5.1:exec (default-cli) @ BankAccount ---
    Account Number: 0000
    Account Holder: Unknown
    Balance: $0.0
    -----
    Account Number: ACC-001
    Account Holder: Alice
    Balance: $0.0
    -----
    Account Number: ACC-002
    Account Holder: Bob
    Balance: $5000.75
    -----
    -----
    BUILD SUCCESS
    -----
    Total time: 1.172 s
    Finished at: 2026-02-22T15:10:23+05:00
    -----
    -----
    -----
```

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Task # 2:

```

/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to view the code snippet.
 */

package com.mycompany.point;

/**
 *
 * @author rehan
 */
public class Point {
    private int x;
    private int y;

    // Default constructor
    public Point() {
        this.x = 0;
        this.y = 0;
    }

    public void displayInfo() {
        System.out.println("Point(" + x + ", " + y + ")");
    }

    public static void main(String[] args) {
        Point p = new Point();
        p.displayInfo();
    }
}

```

OutPut:

```
cd /home/rehan/NetBeansProjects/point; JAVA_HOME=/usr/lib/jv
WARNING: A terminally deprecated method in sun.misc.Unsafe h
WARNING: sun.misc.Unsafe::staticFieldBase has been called by
WARNING: Please consider reporting this to the maintainers o
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in
Scanning for projects...

-----< com.mycompany:point >-----
[Building point 1.0-SNAPSHOT
  from pom.xml]
-----[ jar ]-----

--- resources:3.3.1:resources (default-resources) @ point ---
skip non existing resourceDirectory /home/rehan/NetBeansProj

--- compiler:3.13.0:compile (default-compile) @ point ---
Recompiling the module because of changed source code.
Compiling 1 source file with javac [debug release 25] to tar

--- exec:3.5.1:exec (default-cli) @ point ---
Point(0, 0)

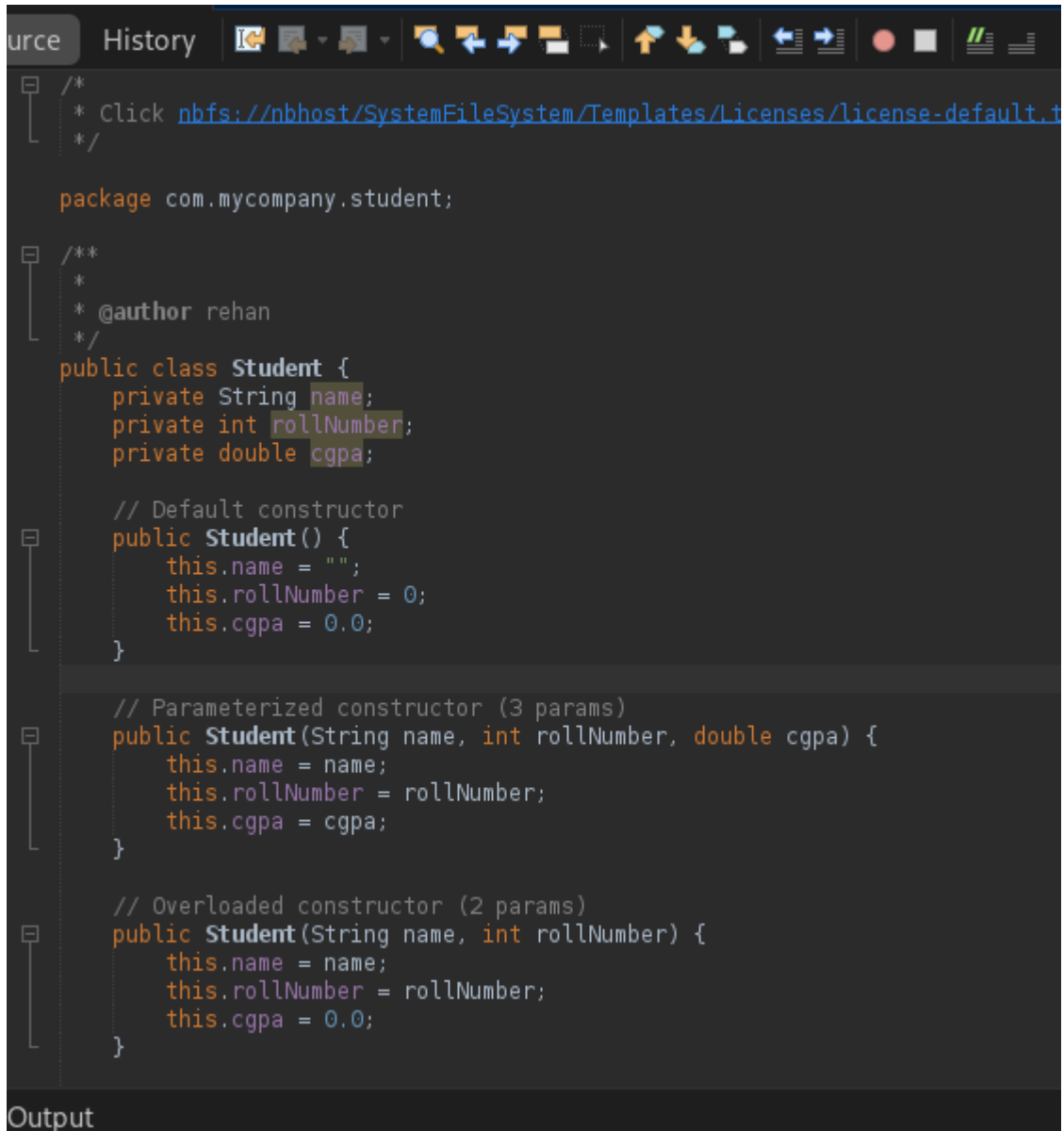
BUILD SUCCESS

Total time: 2.851 s
Finished at: 2026-02-22T15:18:14+05:00
```

Output

Transferring Maven repository

Task # 3



```
source History   
/*  
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.t  
 */  
  
package com.mycompany.student;  
  
/**  
 *  
 * @author rehan  
 */  
public class Student {  
    private String name;  
    private int rollNumber;  
    private double cgpa;  
  
    // Default constructor  
    public Student() {  
        this.name = "";  
        this.rollNumber = 0;  
        this.cgpa = 0.0;  
    }  
  
    // Parameterized constructor (3 params)  
    public Student(String name, int rollNumber, double cgpa) {  
        this.name = name;  
        this.rollNumber = rollNumber;  
        this.cgpa = cgpa;  
    }  
  
    // Overloaded constructor (2 params)  
    public Student(String name, int rollNumber) {  
        this.name = name;  
        this.rollNumber = rollNumber;  
        this.cgpa = 0.0;  
    }  
}
```

Output

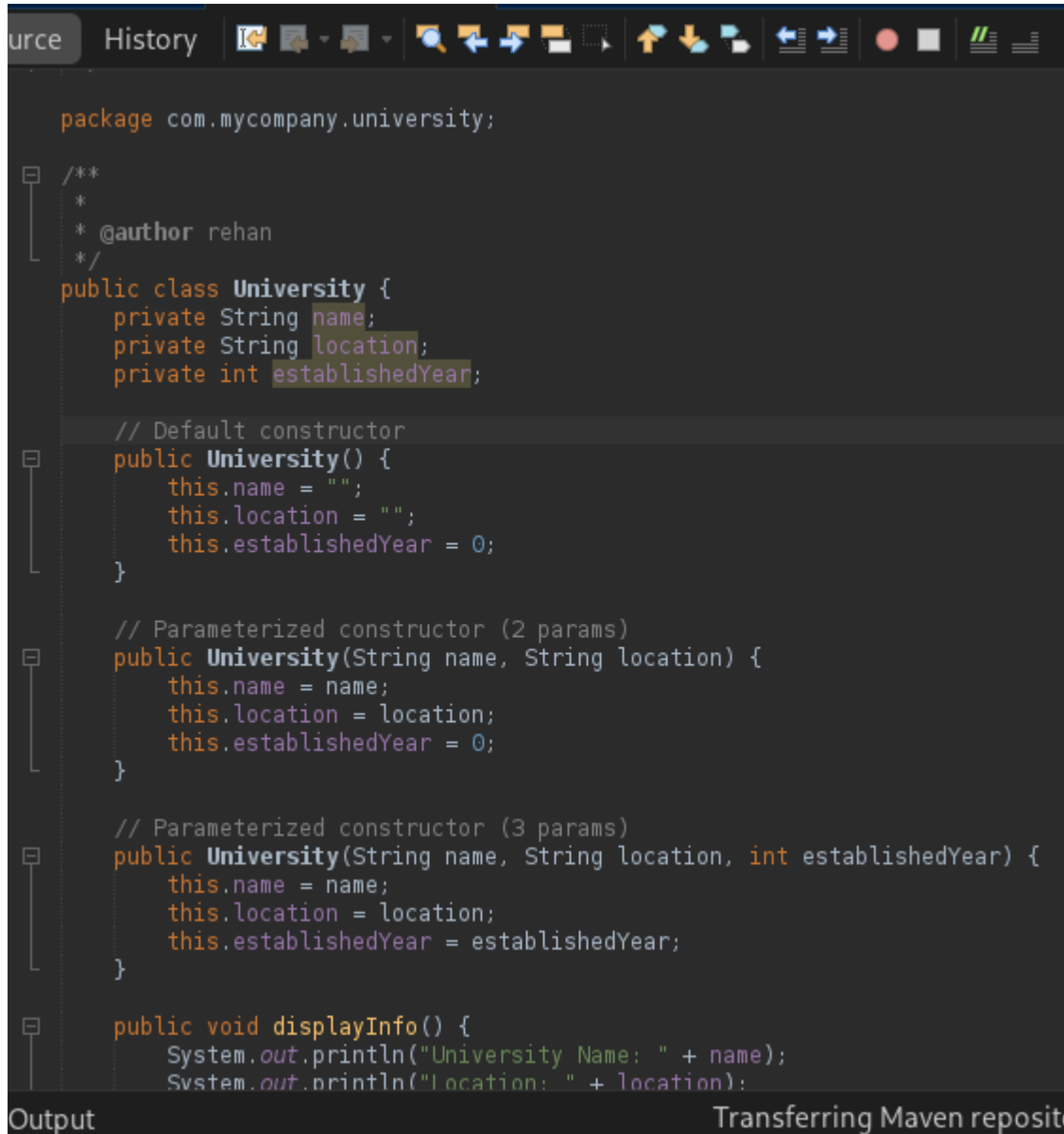
Output:

```
Building Student 1.0-SNAPSHOT
  from pom.xml
-----[ jar ]-----
--- resources:3.3.1:resources (default-resources) @ Student ---
- skip non existing resourceDirectory /home/rehan/NetBeansProj
--- compiler:3.13.0:compile (default-compile) @ Student ---
- Recompiling the module because of changed source code.
- Compiling 1 source file with javac [debug release 25] to tar
--- exec:3.5.1:exec (default-cli) @ Student ---
Name:
Roll Number: 0
CGPA: 0.0
-----
Name: Rehan
Roll Number: 101
CGPA: 3.8
-----
Name: Abdullah
Roll Number: 102
CGPA: 0.0
-----
-----
BUILD SUCCESS
-----
Total time: 2.905 s
Finished at: 2026-02-22T15:28:27+05:00
-----
```

Output

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Task # 4:



The image shows a screenshot of an IDE with a dark theme. The main editor window displays Java code for a `University` class. The code includes package declarations, a Javadoc comment, private fields for `name`, `location`, and `establishedYear`, and three constructors: a default constructor, a two-parameter constructor, and a three-parameter constructor. A `displayInfo` method is also present. The IDE's toolbar is visible at the top, and a sidebar on the left shows a project tree. At the bottom, a terminal window displays the output of a Maven command.

```
package com.mycompany.university;

/**
 *
 * @author rehan
 */
public class University {
    private String name;
    private String location;
    private int establishedYear;

    // Default constructor
    public University() {
        this.name = "";
        this.location = "";
        this.establishedYear = 0;
    }

    // Parameterized constructor (2 params)
    public University(String name, String location) {
        this.name = name;
        this.location = location;
        this.establishedYear = 0;
    }

    // Parameterized constructor (3 params)
    public University(String name, String location, int establishedYear) {
        this.name = name;
        this.location = location;
        this.establishedYear = establishedYear;
    }

    public void displayInfo() {
        System.out.println("University Name: " + name);
        System.out.println("Location: " + location);
    }
}
```

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OutPut:

```
[-] Building University 1.0-SNAPSHOT
    from pom.xml
-----[ jar ]-----
[-] --- resources:3.3.1:resources (default-resources) @ University ---
- skip non existing resourceDirectory /home/rehan/NetBeansProj
[-] --- compiler:3.13.0:compile (default-compile) @ University ---
- Recompiling the module because of changed source code.
- Compiling 1 source file with javac [debug release 25] to tar
[-] --- exec:3.5.1:exec (default-cli) @ University ---
University Name:
Location:
Established Year: 0
-----
University Name: DUET
Location: NED
Established Year: 0
-----
University Name: KU
Location: UET
Established Year: 1986
-----
-----
BUILD SUCCESS
-----
Total time: 2.969 s
Finished at: 2026-02-22T15:33:41+05:00
-----
```

Output

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